Iteration

17 September 2026 12:48

```
for value in range(start, end):
    #code
```

[end-1]

For example:

```
start = int(input('Enter the start value: '))
end = int(input('Enter the end value: '))
for i in range(start, end+1):
    if i % 3 == 0:
        print(i, end=' ')
```

Output:

```
Enter the start value: 10
Enter the end value: 30
12 15 18 21 24 27 30
```

skip value / Jump value

```
for i in range(2, 100, 2):
    print(i, end = ', ')
```

Output:

```
2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38,
40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74,
76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98,
```

not included in the result

```
for i in range(100, 2, -2):
    print(i, end = ', ')
```

Output:

```
100, 98, 96, 94, 92, 90, 88, 86, 84, 82, 80, 78, 76, 74, 72, 70, 68, 66,
64, 62, 60, 58, 56, 54, 52, 50, 48, 46, 44, 42, 40, 38, 36, 34, 32, 30,
28, 26, 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4,
```

```
# Code to find the sum of each digit, Number is entered by the user.
num = int(input('Enter any positive number: '))
if num > 0:
```

```
        sum = 0
        while num > 0:
            sum = sum + num % 10
            num = num // 10 # floor division
        print(f'Sum of digits: {sum}')
    else:
        print(f'{num} is negative or zero')
```

body of the loop

quotient

Outputs:

1. Enter any positive number: 628
   Sum of digits: 16

2. Enter any positive number: -628
   -628 is negative or zero

While loop in Python

```
while <condition>:
    Statement(s)
```

The loop will continue till the condition is satisfied.

Example of infinite while loop:

```
while True:
    print('I am infinite!')

    i = input('Press 0 to exit:')
    if i == '0':
        break # It breaks or terminates the current loop.
```

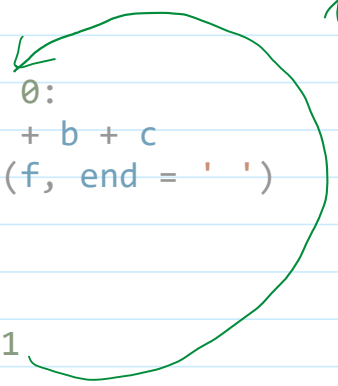
Output:

```
Enter lower bound: 10
Enter upper bound: 50
11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49
Number of Odd values: 20
PS C:\Users\bluej\Documents\Students\AaravSharmaCBSE2628\Git\Coding> &
udents/AaravSharmaCBSE2628/Git/Coding/Chapter0/whileDemoTwo.py
Enter lower bound: 50
Enter upper bound: 10
upper bound value is less than lower bound
```

```
N = int(input('Enter number of terms: '))
a = -1
b = 0
```

0 1 2 3 4

```
a = -1
b = 0
c = 1
while N > 0:
    f = a + b + c
    print(f, end = ' ')
    a = b
    b = c
    c = f
    N -= 1
```



0 1 2 3 6

a = -1 + 0 + 1 + 2
b = 0 + 1 + 2 + 3
c = -1 + 2 + 3 + 6
f = 0 1 2 3 6